

06 — AI & Scraping Pipeline

Official-document ingestion, OCR, deduplication, AI summaries and human review

CampusFlow • Production Specification • v1.0 • October 2026

1. Source-of-Truth Rule

AI is assistive, not authoritative. The original official notice/document remains the source of truth.

Every AI summary displays: AI-generated summary; refer to the official notice for complete and authoritative information.

2. Ingestion Pipeline

```
Official website / Admin upload
↓
Validation → SHA-256 → duplicate check
↓
PDF extraction / OCR
↓
Raw source text
↓
Structured extraction
↓
Human review
↓
Publish
↓
Async AI summary
```

3. Scraper Service

Planned file: backend/src/services/scraper.service.ts.

- Controlled schedule, planned every 30 minutes.
- Fetch official notice/news pages.
- Cheerio for HTML; Playwright only when required.
- Download PDF links.
- Compute SHA-256; skip known hashes.
- pdf-parse for text PDFs.
- tesseract.js for scanned/image documents.
- Store sourceUrl and raw text.
- Respect site access rules and stability.

4. Summary Service

Planned file: backend/src/services/summary.service.ts.

Extract JSON:
{what_changed, who_is_affected, required_action, deadline, audience_hint}
Rules: NEVER invent dates, requirements or penalties. Return null when absent. Maximum four bullets. JSON only.

Validate with Zod. Check that any deadline is explicitly supported by source text. If validation fails, discard summary and keep original notice.

5. OCR / Timetable Rule

Timetable OCR must not publish automatically. Extracted schedule data requires authorized human review before it becomes official CampusFlow data.

6. Example

The supplied academic calendar states CIE-I for 29/10/2026–06/11/2026 and applies to FY MCA and FY MBA. A summary may repeat those facts, but must not invent an action absent from the source.

7. Missing AI Key

AI_PROVIDER_KEY is optional locally. If absent, use a clearly marked mock path and log a server warning. Never expose the key to the frontend.

8. Future Phase 4

Personalized ranking.

Smart reminders 24h before relevant deadlines.

Semantic search using pgvector if justified.

No automated high-impact academic decisions.

9. Official Website Integration

The current PES Modern College of Engineering site exposes Notice Board, News Announcement and event information. These are candidate sources for controlled ingestion; implementation must respect access rules and site stability.